

- 1 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-026 on the 'german_credit' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- german_credit has 1000 source rows, 20 cached features, and a training positive rate of 0.300.\n\nRisks / caveats flagged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-026", "dataset": "german_credit", "stage": "data_understanding"}}
- 2 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-023 on the 'credit_default' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the raw feature recipe for the next model screen on credit_default; it achieved 0.7317 mean training-CV ROC-AUC with about 23 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-023", "dataset": "credit_default", "stage": "feature_engineering"}}
- 3 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'leakage' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (strong empirical confidence): A large safe-versus-unsafe score gap is a severity demonstration, not the primary proof of leakage.\n\nWhy it matters: Bank marketing and online shoppers showed +0.1328 to +0.1575 registry gaps, while semantics identified the forbidden fields.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R06", "category": "leakage"}}
- 4 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'knowledge'. What would that look like in practice, and how do I che

stant", "content": "Other findings:\n- Use the raw feature recipe for the next model screen on adult; it achieved 0.8955 mean training-CV ROC-AUC with about 14 feature s.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-003", "dataset": "adult", "stage": "feature_engineering"}}

9 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'evaluation'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Consume the final holdout once, after all choices are locked, and never tune from its result."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R17", "category": "evaluation"}}

10 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'optimization' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Compare tuning and feature changes with a paired baseline on the same folds and demand a predeclared useful margin.\n\nWhy it matters: The campaign used a 0.001 CV gate; this is a project convention, not a universal constant.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds.\n\nExceptions: Use a larger gate when business value, noise or retraining cost demands it."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R16", "category": "optimization"}}

11 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-015 on the 'breast_cancer' dataset find during the 'optimization reliability' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The preselected logistic_regression/baseline recipe scored 0.9960 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.9858-1.0000).\n\nOther findings:\n- Do not promote features from importance alone; require decision-time availability, cross-fold stability, production availability, and monitoring."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-015", "dataset": "breast_cancer", "stage": "optimization_reliability"}}

12 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-007 on the 'bank_marketing' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for bank_marketing: ['duration']; their apparent training-CV AUC lift was +0.1842.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-007", "dataset": "bank_marketing", "stage": "leakage_audit"}}

13 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-019 on the 'heart_disease' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for heart_disease using raw features (0.8978±0.0244 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-019", "dataset": "heart_disease", "stage": "model_selection"}}

14 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'model selection' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Use mean performance, fold stability, probability quality, runtime and operational constraints-not ROC-AUC alone.\n\nWhy it matters: Small AUC wins can coincide with worse log loss, calibration, recall or latency.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R14", "category": "model_selection"}}

15 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-006 on the 'bank_marketing' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- bank_marketing has 45211 source rows, 16 cached features, and a training positive rate of 0.117.\n\nRisks / caveats fla

gged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-006", "dataset": "bank_marketing", "stage": "data_understanding"}}

16 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'leakage' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): A suspicious name or high univariate AUC starts a semantic review; it does not prove leakage.\n\nWhy it matters: Availability requires source lineage and timestamps; useful real predictors can also be highly associated.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R05", "category": "leakage"}}

17 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'categoricals'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Preserve raw categorical semantics and fit encoders inside folds; do not treat arbitrary category codes as ordered numbers."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R11", "category": "categoricals"}}

18 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-010 on the 'bank_marketing' dataset find during the 'optimization reliability' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The preselected extra_trees/extra_trees_C02 recipe scored 0.7718 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.7126-0.8290).\n\nOther findings:\n- Do not promote features from importance alone; require decision-time availability, cross-fold stability, production availability, and monitoring."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-010", "dataset": "bank_marketing", "stage": "optimization_reliability"}}

19 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building f

framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'evaluation' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Consume the final holdout once, after all choices are locked, and never tune from its result.\n\nWhy it matters: Repeated consultation converts a holdout into development data.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R17", "category": "evaluation"}}

20 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-030 on the 'german_credit' dataset find during the 'optimization reliability' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The preselected extra_trees/extra_trees_C01 recipe scored 0.7943 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.7180-0.8621).\n\nOther findings:\n- Do not promote features from importance alone; require decision-time availability, cross-fold stability, production availability, and monitoring."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-030", "dataset": "german_credit", "stage": "optimization_reliability"}}

21 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'generalization'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Use temporal, geographic, source-system and out-of-domain slices when those shifts exist."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R19", "category": "generalization"}}

22 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Prediction contract' stage of model building?"}, {"role": "assistant", "content": "The 'Prediction contract' stage (WF-01) follows this flow:\n1. define entity and decision\n2. define prediction moment\n3. define target window\n4. define action and costs\n5. freeze allowable information"}], "metadata": {"source": "workflow_blocks", "id": "WF-01"}}

23 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-029 on the 'german_credit' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for german_credit using interactions features (0.7769 ± 0.0298 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-029", "dataset": "german_credit", "stage": "model_selection"}}

24 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-043 on the 'online_shoppers' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the raw feature recipe for the next model screen on online_shoppers; it achieved 0.7156 mean training-CV ROC-AUC with about 16 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-043", "dataset": "online_shoppers", "stage": "feature_engineering"}}

25 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Algorithm screen' stage of model building?"}, {"role": "assistant", "content": "The 'Algorithm screen' stage (WF-07) follows this flow:\n1. dummy floor\n2. regularized linear model\n3. bagged trees\n4. boosting families\n5. domain-appropriate specialist\n6. rank on identical folds"}], "metadata": {"source": "workflow_blocks", "id": "WF-07"}}

26 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-009 on the 'bank_marketing' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for bank_marketing using ratios features (0.7288 ± 0.0295 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-009", "dataset": "bank_marketing", "stage": "model_selection"}}

27 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Source and lineage gate' stage of model building?"}, {"role": "assistant", "content": "The 'Source and lineage gate' stage (WF-02) follows this flow:\n1. inventory sources\n2. hash immutable snapshot\n3. record event timestamps\n4. map raw-to-modeled fields\n5. label owner and SLA"}], "metadata": {"source": "workflow_blocks", "id": "WF-02"}}

28 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Feature ladder' stage of model building?"}, {"role": "assistant", "content": "The 'Feature ladder' stage (WF-06) follows this flow:\n1. raw baseline\n2. single domain transform\n3. ratios/differences\n4. bounded interactions\n5. external point-in-time features\n6. retain only reproducible lift"}], "metadata": {"source": "workflow_blocks", "id": "WF-06"}}

29 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Train-only EDA' stage of model building?"}, {"role": "assistant", "content": "The 'Train-only EDA' stage (WF-04) follows this flow:\n1. schema and type audit\n2. target and missingness profile\n3. duplicates/entities\n4. distribution slices\n5. association hypotheses without promotion"}], "metadata": {"source": "workflow_blocks", "id": "WF-04"}}

30 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-049 on the 'wine_quality' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for wine_quality using ratios features (0.8609±0.0036 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-049", "dataset": "wine_quality", "stage": "model_selection"}}

31 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-002

on the 'adult' dataset find during the 'leakage audit' stage?}], {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for adult: none.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-002", "dataset": "adult", "stage": "leakage_audit"}}}

32 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-035 on the 'mushroom' dataset find during the 'optimization reliability' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The preselected extra_trees/baseline recipe scored 1.0000 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 1.0000-1.0000).\n\nOther findings:\n- Do not promote features from importance alone; require decision-time availability, cross-fold stability, production availability, and monitoring."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-035", "dataset": "mushroom", "stage": "optimization_reliability"}}}

33 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'feature engineering' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (moderate empirical confidence): Begin with a strong raw baseline and promote the smallest defensible feature set within a predeclared tolerance.\n\nWhy it matters: Seven of ten campaign datasets selected raw features; extra features often added dilution without repeatable lift.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds.\n\nExceptions: Domain transformations may still be essential when raw representation hides a known relationship."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R07", "category": "feature_engineering"}}}

34 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'generalization' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Use temporal, geographic, source-system and out-of-domain slices when those shifts exist.\n\nWhy it matters: Random CV estimates interpolation under the historical mixture, not future operational performance.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input s

imulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R19", "category": "generalization"}}}

35 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'missingness'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Model missingness as an operational state: measure coverage, add indicators when justified, simulate absence, and define fallback behavior."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R12", "category": "missingness"}}}

36 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'llm governance'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Let LLM agents propose hypotheses, challenge semantics and curate cited lessons; deterministic executors own splits, metrics and artifacts."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R20", "category": "llm_governance"}}}

37 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'knowledge' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Store claims with scope, evidence IDs, counterevidence, uncertainty and the next discriminating test.\n\nWhy it matters: A score without context cannot teach a future person or model when the lesson applies.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R21", "category": "knowledge"}}}

38 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a con

crete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'feature reliability' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Importance is only a nomination signal; production reliability additionally requires availability, fold stability, coverage, missing-input resilience and drift monitoring.\n\nWhy it matters: The campaign explicitly rejects importance-only promotion and the agentic worker measures missing-input stress.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R10", "category": "feature_reliability"}}

39 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-025 on the 'credit_default' dataset find during the 'optimization reliability' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The preselected extra_trees/extra_trees_C02 recipe scored 0.7746 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.7326-0.8150).\n\nOther findings:\n- Do not promote features from importance alone; require decision-time availability, cross-fold stability, production availability, and monitoring."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-025", "dataset": "credit_default", "stage": "optimization_reliability"}}

40 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'eda'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Fit target-aware EDA, imputers, encoders, scalers, selectors and learned aggregations on training partitions only."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R03", "category": "eda"}}

41 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'model selection' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (strong empirical confidence): There is no universal best tabular algorithm; screen a transparent linear baseline and several nonlinear families on identical folds.\n\nWhy it matters: LightGBM had best mean rank across the registry, ExtraTrees won 7/10 in the controlled campaign

gn, logistic won churn, and an ensemble won HyperAck.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R13", "category": "model_selection"}}

42 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'leakage'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Treat post-outcome, future, target-derived, cross-split aggregate, duplicate/entity and preprocessing leakage as separate failure modes."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R04", "category": "leakage"}}

43 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'evaluation' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Choose metrics and thresholds from the real decision cost; report ranking, probability quality and operating-point errors together.\n\nWhy it matters: AUC does not specify calibration or the false-positive/false-negative tradeoff.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R18", "category": "evaluation"}}

44 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-031 on the 'mushroom' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- mushroom has 8124 source rows, 22 cached features, and a training positive rate of 0.482.\n\nRisks / caveats flagged:\n- 1 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-031", "dataset": "mushroom", "stage": "data_understanding"}}

45 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Split design' stage of model building?"}, {"role": "assistant", "content": "The 'Split design' stage (WF-03) follows this flow:\n1. identify independence unit\n2. choose group/time/domain split\n3. lock test IDs\n4. verify no overlap\n5. version split manifest"}], "metadata": {"source": "workflow_blocks", "id": "WF-03"}}

46 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'splitting' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Split by the real independence unit and time direction before preprocessing or feature selection.\n\nWhy it matters: Random rows leak entity history and future regimes when observations are related.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R02", "category": "splitting"}}

47 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Conservative optimization' stage of model building?"}, {"role": "assistant", "content": "The 'Conservative optimization' stage (WF-08) follows this flow:\n1. freeze objective\n2. small explicit search\n3. paired fold comparison\n4. proper scoring and runtime\n5. accept or keep baseline"}], "metadata": {"source": "workflow_blocks", "id": "WF-08"}}

48 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'feature engineering' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Every generated feature needs explicit lineage, availability time, units, null behavior and a reproducible formula.\n\nWhy it matters: A mathematically valid feature can still be unavailable, unstable or impossible to reproduce online.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test

it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R09", "category": "feature_engineering"}}

49 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'llm governance' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high system-design confidence): Let LLM agents propose hypotheses, challenge semantics and curate cited lessons; deterministic executors own splits, metrics and artifacts.\n\nWhy it matters: This preserves creativity without letting generated text become measurement.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R20", "category": "llm_governance"}}

50 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-008 on the 'bank_marketing' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the ratios feature recipe for the next model screen on bank_marketing; it achieved 0.7172 mean training-CV ROC-AUC with about 38 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-008", "dataset": "bank_marketing", "stage": "feature_engineering"}}}

51 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-027 on the 'german_credit' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for german_credit: none.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-027", "dataset": "german_credit", "stage": "leakage_audit"}}}

52 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'missingness' stage of building a tabular ML model, what does evidence-based practice recommend, and why

does it matter?}], {"role": "assistant", "content": "Rule (high methodological confidence): Model missingness as an operational state: measure coverage, add indicators when justified, simulate absence, and define fallback behavior.\n\nWhy it matters: A feature that disappears in production can dominate offline ranking yet make the system brittle.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R12", "category": "missingness"}}}

53 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'leakage' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Treat post-outcome, future, target-derived, cross-split aggregate, duplicate/entity and preprocessing leakage as separate failure modes.\n\nWhy it matters: One correlation detector cannot identify all leakage mechanisms.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R04", "category": "leakage"}}}

54 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Leakage audit' stage of model building?"}, {"role": "assistant", "content": "The 'Leakage audit' stage (WF-05) follows this flow:\n1. semantic availability matrix\n2. lineage/timestamp check\n3. target-proxy review\n4. split/preprocessing audit\n5. safe-vs-unsafe severity ablation"}], "metadata": {"source": "workflow_blocks", "id": "WF-05"}}}

55 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-040 on the 'spambase' dataset find during the 'optimization reliability' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The preselected hist_gradient_boosting/baseline recipe scored 0.9849 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.9752-0.9925).\n\nOther findings:\n- Do not promote features from importance alone; require decision-time availability, cross-fold stability, production availability, and monitoring."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-040", "dataset": "spambase", "stage": "optimization_reliability"}}}

56 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-042 on the 'online_shoppers' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for online_shoppers: ['PageValues']; their apparent training-CV AUC lift was +0.1968.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-042", "dataset": "online_shoppers", "stage": "leakage_audit"}}

57 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-017 on the 'heart_disease' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for heart_disease: none.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-017", "dataset": "heart_disease", "stage": "leakage_audit"}}

58 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-032 on the 'mushroom' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for mushroom: none.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-032", "dataset": "mushroom", "stage": "leakage_audit"}}

59 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'feature reliability'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Under

lying rule: Importance is only a nomination signal; production reliability additionally requires availability, fold stability, coverage, missing-input resilience and drift monitoring."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R10", "category": "feature_reliability"}}

60 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-021 on the 'credit_default' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- credit_default has 30000 source rows, 23 cached features, and a training positive rate of 0.221.\n\nRisks / caveats flagged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-021", "dataset": "credit_default", "stage": "data_understanding"}}

61 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'promotion'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: A candidate is not production-ready until provenance, independent confirmation, feature contracts, monitoring, fairness/privacy and rollback gates are reviewed."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R22", "category": "promotion"}}

62 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-036 on the 'spambase' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- spambase has 4601 source rows, 57 cached features, and a training positive rate of 0.394.\n\nRisks / caveats flagged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-036", "dataset": "spambase", "stage": "data_understanding"}}

63 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'categoricals' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Preserve raw categorical semantics and fit encoders inside folds; do not treat arbitrary category codes as ordered numbers.\n\nWhy it matters: Legacy factorized caches lose labels and can introduce false ordinal structure.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R11", "category": "categoricals"}}

64 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'problem contract' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Write the prediction moment, target window, action, unit of prediction and error costs before inspecting target associations.\n\nWhy it matters: Leakage is defined relative to what is knowable at a specific decision moment, not by a column name.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R01", "category": "problem_contract"}}

65 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-046 on the 'wine_quality' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- wine_quality has 6497 source rows, 11 cached features, and a training positive rate of 0.633.\n\nRisks / caveats flagged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-046", "dataset": "wine_quality", "stage": "data_understanding"}}

66 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a con

crete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-004 on the 'adult' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- lightgbm is the training-CV model candidate for adult using raw features (0.8955±0.0065 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-004", "dataset": "adult", "stage": "model_selection"}}

67 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-044 on the 'online_shoppers' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for online_shoppers using raw features (0.7366±0.0072 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-044", "dataset": "online_shoppers", "stage": "model_selection"}}

68 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What is the reusable workflow for the 'Knowledge capture' stage of model building?"}, {"role": "assistant", "content": "The 'Knowledge capture' stage (WF-10) follows this flow:\n1. observed result\n2. narrow interpretation\n3. limitations/counterexample\n4. evidence paths and hashes\n5. next discriminating test\n6. human review status"}], "metadata": {"source": "workflow_blocks", "id": "WF-10"}}

69 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-039 on the 'spambase' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- hist_gradient_boosting is the training-CV model candidate for spambase using raw features (0.9874±0.0021 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-039", "dataset": "spambase", "stage": "model_selection"}}

70 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cu

ting corners on 'model selection'. What would that look like in practice, and how do I check?}], {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: There is no universal best tabular algorithm; screen a transparent linear baseline and several nonlinear families on identical folds."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R13", "category": "model_selection"}}}

71 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-018 on the 'heart_disease' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the raw feature recipe for the next model screen on heart_disease; it achieved 0.8700 mean training-CV ROC-AUC with about 13 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-018", "dataset": "heart_disease", "stage": "feature_engineering"}}

72 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-034 on the 'mushroom' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for mushroom using raw features (1.0000±0.0000 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-034", "dataset": "mushroom", "stage": "model_selection"}}

73 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-024 on the 'credit_default' dataset find during the 'model selection' stage?"}, {"role": "assistant", "content": "Other findings:\n- extra_trees is the training-CV model candidate for credit_default using raw features (0.7363±0.0159 ROC-AUC across folds).\n\nKnown limitations:\n- A cross-validation leader is not a universal algorithm winner and has not yet seen the locked holdout."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-024", "dataset": "credit_default", "stage": "model_selection"}}

74 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest

st next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-011 on the 'breast_cancer' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- breast_cancer has 569 source rows, 30 cached features, and a training positive rate of 0.374.\n\nRisks / caveats flagged:\n- 1 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-011", "dataset": "breast_cancer", "stage": "data_understanding"}}

75 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'optimization'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Optimize only after data, leakage, features, metric and candidate family are locked."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R15", "category": "optimization"}}

76 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'model selection'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Use mean performance, fold stability, probability quality, runtime and operational constraints-not ROC-AUC alone."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R14", "category": "model_selection"}}

77 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-038 on the 'spambase' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the raw feature recipe for the next model screen on spambase; it achieved 0.9869 mean training-CV ROC-AUC with about 57 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-038", "dataset": "spambase", "stage": "feature_engineering"}}

78 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-048 on the 'wine_quality' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the ratios feature recipe for the next model screen on wine_quality; it achieved 0.8429 mean training-CV ROC-AUC with about 32 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-048", "dataset": "wine_quality", "stage": "feature_engineering"}}

79 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'eda' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Fit target-aware EDA, imputers, encoders, scalers, selectors and learned aggregations on training partitions only.\n\nWhy it matters: Any statistic that sees evaluation labels or distributions can make validation optimistic.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R03", "category": "eda"}}

80 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-001 on the 'adult' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- adult has 48842 source rows, 14 cached features, and a training positive rate of 0.239.\n\nRisks / caveats flagged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-001", "dataset": "adult", "stage": "data_understanding"}}

81 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-022 on the 'credit_default' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the ratios feature recipe for the next model screen on credit_default; it achieved 0.8429 mean training-CV ROC-AUC with about 32 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-022", "dataset": "credit_default", "stage": "leakage_audit"}}

ssistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for credit_default: none.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-022", "dataset": "credit_default", "stage": "leakage_audit"}}

82 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'leakage'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: A large safe-versus-unsafe score gap is a severity demonstration, not the primary proof of leakage."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R06", "category": "leakage"}}

83 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-016 on the 'heart_disease' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- heart_disease has 303 source rows, 13 cached features, and a training positive rate of 0.459.\n\nRisks / caveats flagged:\n- 3 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-016", "dataset": "heart_disease", "stage": "data_understanding"}}

84 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-033 on the 'mushroom' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the raw feature recipe for the next model screen on mushroom; it achieved 1.0000 mean training-CV ROC-AUC with about 22 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-033", "dataset": "mushroom", "stage": "feature_engineering"}}

85 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building f

framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'feature engineering'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Begin with a strong raw baseline and promote the smallest defensible feature set within a predeclared tolerance."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R07", "category": "feature_engineering"}}

86 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-047 on the 'wine_quality' dataset find during the 'leakage audit' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- The leakage detector caught its synthetic exact-target canary: True.\n\nOther findings:\n- Declared decision-time exclusions for wine_quality: none.\n\nKnown limitations:\n- Name, univariate AUC, and uniqueness rules propose review; only decision-time semantics can confirm leakage."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-047", "dataset": "wine_quality", "stage": "leakage_audit"}}

87 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'feature engineering'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Every generated feature needs explicit lineage, availability time, units, null behavior and a reproducible formula."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R09", "category": "feature_engineering"}}

88 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'optimization' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (high methodological confidence): Optimize only after data, leakage, features, metric and candidate family are locked.\n\nWhy it matters: Early broad tuning increases adaptive overfitting and obscures what caused improvement.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical fo

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lds.}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R15", "category": "optimization"}}
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89 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'optimization'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Compare tuning and feature changes with a paired baseline on the same folds and demand a predeclared useful margin."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R16", "category": "optimization"}}
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90 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'evaluation'. What would that look like in practice, and how do I check?"}, {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Choose metrics and thresholds from the real decision cost; report ranking, probability quality and operating-point errors together."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R18", "category": "evaluation"}}
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91 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-041 on the 'online_shoppers' dataset find during the 'data understanding' stage?"}, {"role": "assistant", "content": "Observed evidence:\n- online_shoppers has 12330 source rows, 17 cached features, and a training positive rate of 0.155.\n\nRisks / caveats flagged:\n- 0 features triggered missingness, identifier, near-constant, or random-split shift review rules.\n\nKnown limitations:\n- Random train/holdout PSI is a smoke check, not evidence of temporal production stability.\n- The cached UCI matrices factorized categoricals before this campaign; raw category semantics must be restored for production encoders."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-041", "dataset": "online_shoppers", "stage": "data_understanding"}}
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92 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "I suspect my pipeline is cutting corners on 'splitting'. What would that look like in practice, and how do I che
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ck?}], {"role": "assistant", "content": "The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation. To check it directly: Record the decision and test it on identical folds. Underlying rule: Split by the real independence unit and time direction before preprocessing or feature selection."}], "metadata": {"source": "model_building_rules_diagnostic", "rule_id": "DCLAB-R02", "category": "splitting"}}

93 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "In the 'feature engineering' stage of building a tabular ML model, what does evidence-based practice recommend, and why does it matter?"}, {"role": "assistant", "content": "Rule (moderate empirical confidence): Generate features from domain invariances and mechanisms before blind combinatorial expansion.\n\nWhy it matters: Ratios helped bank marketing and wine quality; interactions helped German credit; churn charge/tenure features improved logistic AUC by +0.0050.\n\nHow you'd know it was violated: The result changes materially under a valid ablation, later-time slice, missing-input simulation, or independent evaluation.\n\nHow to test it: Record the decision and test it on identical folds."}], "metadata": {"source": "model_building_rules", "rule_id": "DCLAB-R08", "category": "feature_engineering"}}

94 {"messages": [{"role": "system", "content": "You are a senior data-science / ML engineering research assistant. You answer using the DCLab evidence-first model-building framework: state the observed evidence, your interpretation and its uncertainty, a concrete decision or recommendation, remaining risks or missing evidence, and the smallest next falsifiable test."}, {"role": "user", "content": "What did experiment EXP-028 on the 'german_credit' dataset find during the 'feature engineering' stage?"}, {"role": "assistant", "content": "Other findings:\n- Use the interactions feature recipe for the next model screen on german_credit; it achieved 0.7601 mean training-CV ROC-AUC with about 56 features.\n\nKnown limitations:\n- Generic mathematical feature generation may lack domain meaning; a domain expert must review production semantics."}], "metadata": {"source": "agent_memory", "experiment_id": "EXP-028", "dataset": "german_credit", "stage": "feature_engineering"}}

95